

Transient histone acetylation dynamics at the Cxcl9–Cxcl11 chemokine locus

Genome browser inspection of mm10 chr5:92.32–92.38 Mb revealed low baseline acetylation across the Cxcl9–Cxcl11 cluster at 0 hr, a pronounced induction at 6 hr, and a partial attenuation by 24 hr that nevertheless remained above baseline for all marks (Figure 1). The 6 hr response was dominated by discrete, high-amplitude H3K27ac summits centered on Cxcl10–Cxcl11 promoter-proximal and nearby intergenic regions, with comparatively weaker H3K27ac near Cxcl9. In contrast, H2BK12ac increased broadly across the interval, and H4K5ac formed a right-skewed cluster overlapping the Cxcl10–Cxcl11 portion of the locus. By 24 hr, all three marks declined from their 6 hr maxima but retained locus-specific elevations—H3K27ac peaks persisted over Cxcl10–Cxcl11, while H2BK12ac and H4K5ac domains narrowed and decreased in amplitude (Figure 1).

These mark- and position-specific dynamics support a transient enhancer/promoter activation centered on Cxcl10–Cxcl11 with concomitant, more distributed chromatin engagement across the locus. Specifically, the sharp H3K27ac peaks at 6 hr at promoter-proximal and putative enhancer elements are characteristic of regulatory element activation (Akagha et al., 2026), whereas the broader H2BK12ac and H4K5ac elevations are consistent with widespread nucleosomal acetylation accompanying transcriptional engagement. The muted H3K27ac response at Cxcl9 relative to Cxcl10–Cxcl11 suggests either weaker or delayed activation of Cxcl9 under these conditions, aligning with evidence that the CXCR3-ligand cassette (Cxcl9/10/11) can exhibit coordinated yet nonredundant and context-dependent regulation (Ridley et al., 2023). The partial retention of all three acetylation signals at 24 hr indicates that the 6 hr surge reflects a transiently heightened regulatory and transcriptional state that resolves but does not fully return to baseline, a temporal pattern typical of inducible H3K27ac-marked enhancers and promoters (Akagha et al., 2026).

Collectively, the time-ordered profiles quantify a mid-time (6 hr) peak of regulatory activity focused on Cxcl10–Cxcl11—narrow H3K27ac summits at their promoters and adjacent enhancers with broader H2BK12ac/H4K5ac spanning the local genomic neighborhood—

followed by partial de-acetylation by 24 hr, consistent with transient enhancer/promoter activation coupled to transcriptional engagement at this chemokine locus (Figure 1).

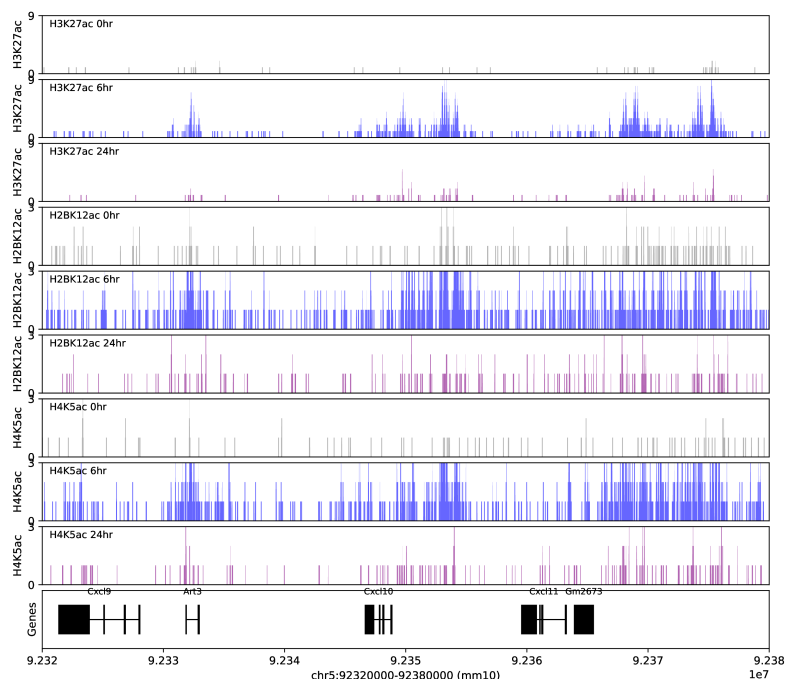


Figure 1. Transient induction of acetylation near Cxcl10–Cxcl11 at 6 hr. Tracks for H3K27ac, H2BK12ac, and H4K5ac at 0, 6, and 24 hr across chr5:92.32–92.38 Mb show minimal baseline signal, a marked increase at 6 hr, and partial decline by 24 hr. The strongest changes occur near the Cxcl10–Cxcl11 region, with weaker responses toward Cxcl9.

References

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